## ■ TEST PROGRAM

### A. HOW TO ENTER THE TEST PROGRAM

1 Set the [HOST SELECT] switch at the [MIDI].
2 Connect the [MIDI IN] to the [MIDI OUT] via a MIDI cable,
3 While pressing the [CURSOR] and [ – /NO] and [ + /YES] switches, turn on the power switch.

The TG100 will indicate that you have entered the Test Program by displaying the following message.

`TG100 TEST V#.##`    `C:AT +:FCT -:EXT`

Use the [CURSOR], [ + /YES] or [ – /NO] switches to select the appropriate test mode. If you press the [CURSOR], the auto test mode will be initiated. If you press the [ + /YES], the TG100 will execute Test 11, ''11. Factory settings'', and then automatically exit the test mode and return to play mode (refer to Test 11 for details). If you press the [ – /NO], you will exit the test mode and return to play mode.

### B. PROCEEDING THROUGH THE TESTS

The TG100 will run the INITIAL TEST routine (refer to the INITIAL TEST section for details) and indicate that you have entered the Test Program by displaying the following message.

`00: TEST NO.01 ?`

Use the [ + /YES] and [ – /NO] switch to move through the various tests of the test program, and then press the [CURSOR] to start the test from that number.

Pressing: [ + /YES] will execute the test which follows the current test.
[ – /NO] will execute the test which precedes the current test.

If you press the [CURSOR] without selecting a test number, the program will start from TEST 1.

**TEST SELECTION WHEN AN ERROR IS DETECTED**

In each test shown below, if an NG (No Good) error is detected, the following operations will make the TG100 retry the test or perform another test.
TEST 3: MIDI TEST
TEST 4: TO HOST TEST

Pressing: [CURSOR] will execute the currently selected test.
[ + /YES] will execute the test which follows the current test.
[ – /NO] will execute Test 12, ''12. EXIT'' (refer to Test 12 for details).

**INITIAL TEST**

A read/write test of EEPROM(IC 4) and PSRAM (IC 3) will be performed automatically when the test program is initiated.

**DISPLAY OF TEST RESULTS**

NG `00: EEPROM NG`

NG `00: PSRAM NG`

**TEST END**

Ends after displaying the results. All RAM data is preserved.
If an error is detected, turn off the power switch to exit the test mode.

### TEST 1. LCD DOTS ON AND OFF TEST

Check that all dots of the LCD blink (ON and OFF). Then, rotate the [CONTRAST] and check that the contrast of the LCD changes continuously.

**TEST END**

Press the [CURSOR] to end the test. The system will proceed to the next test.

### TEST 2. PANEL SWITCH TEST

`02: SW PLAY`

Press panel switches consecutively from the [PLAY] switch to switch [ + /YES], according to the order indicated by the LCD.

`02: SW PART`

(e.g. When checking [PART])

If the switch is OK, a beep will sound and you should proceed to test the next switch. If the wrong switch is pressed, and the error message NG will be displayed and no sound will be heard. At this time, if the correct switch is pressed, you will then be able to proceed to test the next switch. The display will indicate OK, if all switches are good.

**DISPLAY OF TEST RESULTS**

OK `02: SW OK`

NG `02: PLAY NG`

(e.g. When [PLAY] is NG)

**TEST END**

When the switch [ + /YES] is pressed, OK is displayed and the test will end.
If an error is detected when the displayed switch is pressed, turn off the power switch to exit the test mode.

### TEST 3. MIDI TEST

`03: MIDI`

After connecting the [MIDI IN] to the [MIDI OUT] via a MIDI cable, execute the test.

**DISPLAY OF TEST RESULTS**

OK `03: MIDI OK`

NG `03: MIDI NG`

**TEST END**

After displaying the result, the test will end. If NG is detected during the test, refer to section B, ''B. PROCEEDING THROUGH THE TESTS''.

### TEST 4. HOST SELECT TEST

`04: HOST SW xxx`

(where xxx = current [HOST SELECT] switch position)

This test is utilized by the factory and it is not intended for field service use.

**TEST END**

Turn off the power to exit the test mode.

### TEST 5. TO HOST TEST

`05: HOST Tx/Rx`

This test is utilized by the factory and it is not intended for field service use.

**TEST END**

Press the [ + /YES] to end the test. The system will proceed to the next test.

### TEST 6: 1kHz SOUND OUTPUT (LINE OUT L) TEST

`06: 1kHz L`

Insert the appropriate 1/4'' phone plugs into each output jack and check LINE OUT(L), LINE OUT (R), PHONES (L/R) outputs. If necessary, verify the frequency, output waveform and output level of each output using a frequency counter, oscilloscope, AC voltmeter (with 12.47 kHz filter). The volume control must be set at maximum for these checks.

Listed below are the specifications and conditions of each output during this test.
LINE OUT(L): 1kHz ± 1.5Hz, sine wave, distortion 0.6% or less, + 1.0dBm ± 2dB (10k ohm load)
LINE OUT(R): less than – 72dBm
PHONES (L): sine wave, + 5.0dBm ± 2dB (150 ohm load)
PHONES (R): less than – 66dBm

**TEST END**

Press the [CURSOR] to end the test. The sound will stop and the TG100 then executes the next test.

### TEST 7: 1kHz SOUND OUTPUT (LINE OUT R) TEST

`07: 1kHz R`

Insert the appropriate 1/4'' phone plugs into each output jack and check LINE OUT(L), LINE OUT (R), PHONES (L/R) outputs. If necessary, verify the frequency, output waveform and output level of each output using a frequency counter, oscilloscope, AC voltmeter (with 12.47 kHz filter). The volume control must be set at maximum for these checks.

Listed below are the specifications and conditions of each output during this test.
LINE OUT(L): less than – 72dBm
LINE OUT(R): 1kHz ± 1.5Hz, sine wave, distortion 0.6% or less, + 1.0dBm ± 2dB (10k ohm load)
PHONES (L): less than – 66dBm
PHONES (R): sine wave, + 5.0dBm ± 2dB (150 ohm load)

**TEST END**

Press the [CURSOR] to end the test. The sound will stop and the TG100 then executes the next test.

### TEST 8: LDSP/PSRAM TEST

`08: LDSP`

After performing the test 11, FACTORY SET, then execute the test 8.

Insert the appropriate 1/4'' phone plugs into each output jack and check LINE OUT(L) output. If necessary, verify the frequency, output waveform and output level of the output using a frequency counter, oscilloscope, AC voltmeter (with 12.47 kHz filter). The volume control must be set at maximum for these checks.

Listed below is the specification and condition of the output during this test.
When the version of the MAIN ROM is V1.06 (ROM = XK731B00);
LINE OUT(L): 1kHz ± 1.5Hz, sine wave, – 15.0dBm ± 2dB (10k ohm load)
The others (ROM = XK731C00 ~ );
LINE OUT(L): 1kHz ± 1.5Hz, sine wave, – 12dBm ± 2dB (10k ohm load)
* The version is displayed when the test program is initiated (refer to How TO ENTER THE TEST PROGRAM section for details).

**TEST END**

Press the [CURSOR] to end the test. The sound will stop and the TG100 then executes the next test.

### TEST 9: PEAK LED AND INPUT LEVEL CONTROL TEST

`09: INPUT`

Insert the appropriate 1/4'' phone plugs into each output jack and apply a sine wave signal of – 20 dBm, 1kHz to the [AUDIO IN], and check LINE OUT(L), LINE OUT(R), PHONES (L/R) outputs. If necessary, verify the output waveform, output level of each output using an oscilloscope, AC voltmeter (with 12.47 kHz filter). The volume control must be set at maximum for these checks.

Listed below are the specifications and conditions of each output during this test.

1. Apply a sine wave signal of – 20dBm, 1kHz to the [AUDIO IN(L)] and ground the [AUDIO IN(R)] terminal.

1-1. INPUT level control: minimum
LINE OUT(L): – 24.0dBm ± 2dB (10k ohm load)
LINE OUT(R): less than – 72dBm

1-2. INPUT level control: maximum
LINE OUT(L): – 8.5dBm ± 2dB (10k ohm load)
LINE OUT(R): less than – 68dBm

2. Apply a sine wave signal of – 20dBm, 1kHz to the [AUDIO IN(R)] and ground the [AUDIO IN(L)] terminal.

2-1. INPUT level control: minimum
LINE OUT(L): less than – 72dBm
LINE OUT(R): – 24.0dBm ± 2dB (10k ohm load)

2-2. INPUT level control: maximum
LINE OUT(L): less than – 68dBm
LINE OUT(R): – 8.5dBm ± 2dB (10k ohm load)

3. Set the [INPUT] level control at maximum, and apply a sine wave signal of – 12dBm, 1kHz to the [AUDIO IN] terminal and check that the [PEAK] LED will turn on.

**TEST END**

Press the [CURSOR] to end the test. The system will proceed to the next test.

### TEST 10: 28 SOUND GENERATION TEST

`10: 28 CH OUT 00`

Confirm that sine wave signals of channel 1 through 28 are being sent to LINE OUT(L) and LINE OUT(R) every 0.6 seconds using an amplifier and speaker to monitor the signals. While these signals are sounding, the LCD will display the following message:

`10: 28 CH OUT xx`

(where xx = current sound channel)

**TEST END**

Press the [CURSOR] to end the test. The sound will stop and the TG100 then executes the next test.

### TEST 11. FACTORY SET TEST

This test is used to initialize the data to the factory settings.
When this test is executed, the following display will appear.

`11: FACT SET`

If you press the [CURSOR], the factory preset data will be restored, and the TG100 executes Test 12, ''12. Exit'' and then automatically exit the test mode and return to play mode.
If you press the [ – /NO], they will not be restored, and the system will exit the test mode and return to play mode.

**DISPLAY OF TEST RESULTS**

If factory settings are restored.

`11: FACT SET OK`

**TEST END**

The LCD displays the result and the test will end.

### TEST 12. EXIT TEST PROGRAM

If this test is executed, the TG100 will exit the test mode and return to play mode.